COPYRIGHT RESERVED BCA (Sem-I) – Computer

System Architecture (102) Core-II

2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

Objective type Questions

Compulsory

1. Choose the correct answer of the following multiple choice questions : 2×10=20

(i) Simplify the Boolean expression: $A(A+B)$

a) AB

b) $A+B$

c) A

d) B

(ii) A Karnaugh Map for n variables has how many cells?

a) n

b) $2n$

c) n^2

d) 2^n

(iii) While simplifying a K-map, Don't care condition(X) is treated as:

a) Only 0

b) Only 1

c) Both 0 and 1 as needed

D) Neither 0 nor 1

(iv) Choose which of the following is/are SOP form :

a) $(X'+Y)(W'+Z')$

b) $X'Y+W'Z'$

c) Both the above

d) None of these

(v) $F=\prod(2,3,4,5)$ is equivalent to

a) $F=m_2.m_3.m_4.m_5$

b) $F=M_2.M_3.M_4.M_5$

c) $F=M_2+M_3+M_4+M_5$

d) None of these

(vi) How many selection lines in 2^N-to-1 multiplexer :

a) $N+1$

b) N

c) N–1

d) None of these

(vii) Which of the following is not a combinational circuit ?

a) multiplexer

b) decoder

c) encoder

d) flip flop

(viii) In IEEE 754 double precision format, the number of bits for mantissa (fraction) is:

a) 23

b) 32

c) 52

d) 64

(ix) Which DMA transfer mode transfers one word per cycle?

a) Burst mode

b) Cycle stealing

c) Transparent mode

d) Programmed I/O

(x) Direct Memory Access (DMA) allows data transfer between:

a) CPU and I/O device only

b) I/O device and memory without CPU involvement

c) CPU and memory only

d) I/O device and ALU

Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. What is the difference between combinational and sequential circuits?

3. Define universal gate. How to obtain NOR gate using NAND gate.

4. Prove the following expressions by algebraic method

   (a) $X'Z + X'Y + XY'Z + YZ = Z + X'Y$

   (b) $X'Z' + XYZ + XZ' = XY + Z'$

5. Solve the following Boolean expression by K-map:

   $X'Y' + YZ + X'YZ'$

6. Differentiate between DMA Burst Transfer and Cycle Stealing.

7. Write the differences between combinational circuit and sequential circuit.

Section–C

Long Answer type Questions

Answer any **two** questions of the following : 15×2=30

8. Explain De Morgan's Theorems. Prove both theorems using algebraic method.

9. Represent 300.5 using the IEEE 754 standard for 64-bit floating point representation.

10. Using a truth table, design an 8-to-1 multiplexer. Indicate the number of data inputs and selection lines.

11. Write the short notes on the following:

   (a) Flip flop

   (b) Adders

   (c) Instruction format

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